SMART COLLEGE USING ARDUINO

A PROJECT REPORT SUBMITTED FOR PRACTICAL EXAMINATION
DIPLOMA IN ELECTRONICS AND COMMUNICATION ENGINEERING BY
DECE STUDENTS
UNDER THE GUIDENCE OF

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ESC GOVT POLYTECHNIC COLLEGE
NANDYAL-518501

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This to certify that this project report entitled “**SMART COLLEGE ”** is the bonafide work of Mr. ____ Pin: _________________________ student of final year DECE 2022-2023 along with the batchmates submitted in partical fulfillment of requirements for the award of DIPLOMA IN ELECTRONIC & COMMUNICATION ENGINEERGING of S.B.T.E.T A.P during the academic session 2020-2023.

Head of the Department :

Project Guide :

Examiner :

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Whatever one does, whatever one achieves, the first credit goes to the Parents for showing their love and affection, nothing would have been responsible.

A PROJECT REPORT ON SMART COLLAGE USING ARDUINO
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A Comprehensive project report has been submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology
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Under the supervision of

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ABSTRACT

Smart college is an IOT related project .the smart college project consists of four applications that are 'Automatic toilet flush', 'Automatic lights on and off' , 'Automatic watering plants', 'Automatic fire alarm'.

1.Automatic toilet flush :

In schools and colleges mainly from the boys toilets we gets smell because of not using flush properly by using the IR sensor we can over it whenever the sensor detects the person it automatically the flushes the toilet by making water pump ON.

2.Automatic lights ON and OFF:

we can ON and OFF the lights automatically in classroom by using IR sensor .whenever the IR sensor detects the students it ON's the light until the students leave from the classroom.

3.Automatic fire alarm:

Many students died due to fire accidents in schools and colleges by using FLAME sensor we can over come fire accidents when any fire accidents takes place in a classroom the flame in the classroom detects it and it gives an alarm to alert to the students ,teacher etc.

4. Automatic watering plants :

We need a green environment around the college by using SOIL MOISTURE sensor we can automatically water the plants whenever the soil is dried up the sensor automatically ON's the water pump and water's the plants.

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CHAPTER – 1

INTRODUCTION

INTRODUCTION

Nowadays, we have remote controls for appliances and other electronic systems which have made our lives easy. Have you ever wondered about smart college which would give the facility of using appliances in a smart way without any mediator for controlling operation of appliances and other electronic devices? Off-course, Yes! But, are the available options cost effective? If the answer is No, we have found a solution to it.

We have come up with a new system called Arduino based smart college using sensors. This system costs effective and can saves the time of the user. Time is a very valuable thing.

Everybody wants to save time as much as they can. New technologies are being introduced to save our time .To save people's time we are introducing smart college system using arduino & sensors.

This system is controlled on it's own by sensing the presence (based on application we are using for) through sensors. So that works can be done and appliances can be operated without any human interaction

CHAPTER – 2

COMPONENTS AND DESCRIPTION

Hardware requirements:

- Arduino Uno
- Flame sensor
- Soil moisture sensor
- IR sensor
- relay
- DC water pump Motors
- Led
- Buzzer

DESCRIPTION

ARDUINO UNO

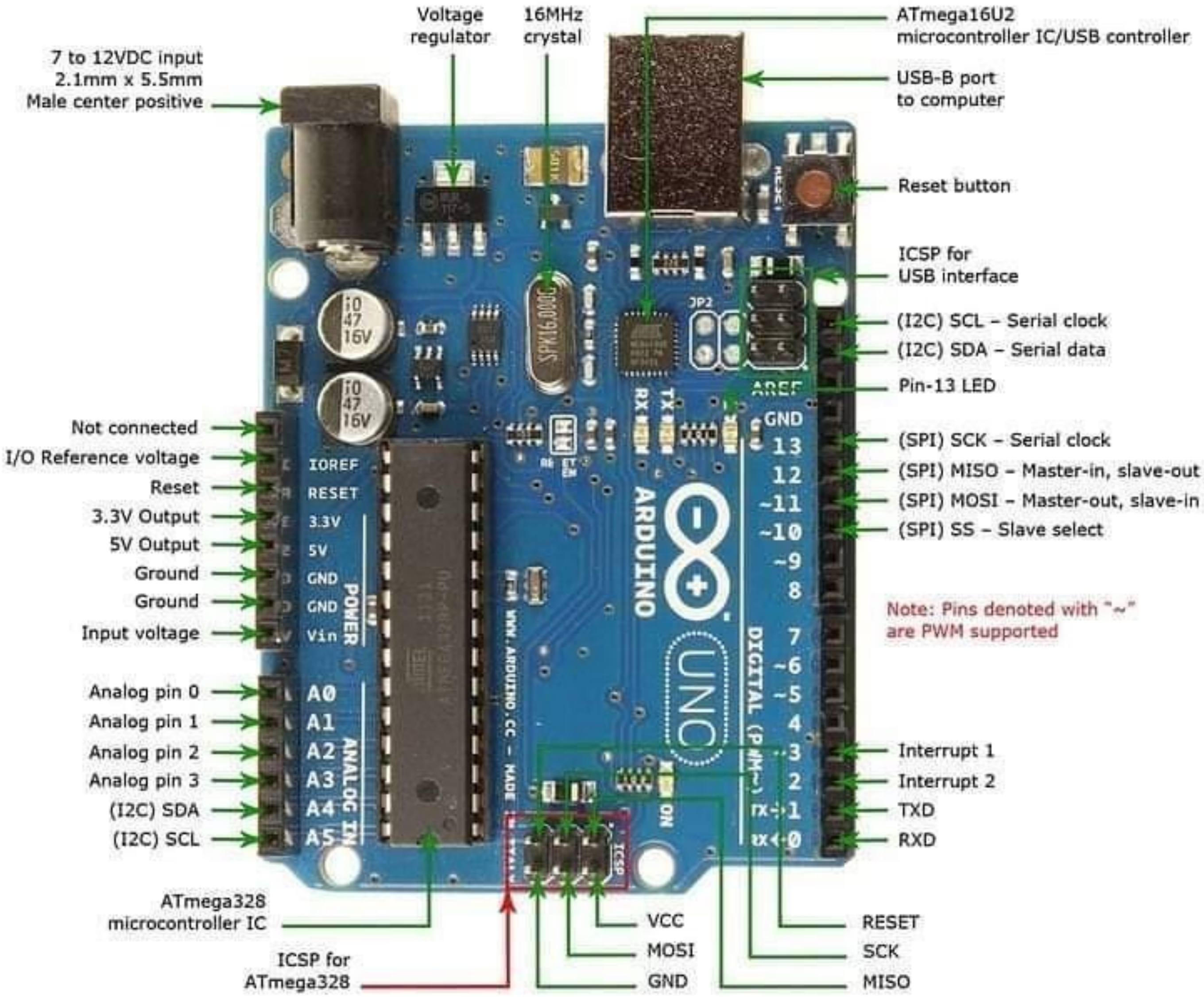
Arduino is an open source computer hardware and software company, project, and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical and digital world.
The project's products are distributed as open-source hardware and software, which are licensed under the
GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and software distribution by anyone.

Arduino boards are available commercially in preassembled form, or as do-ityourself (DIY) kits.
Arduino board designs use a variety of microprocessors and controllers.
The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or Breadboards (shields) and other circuits.

The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers.
The microcontrollers are typically programmed using a dialect of features from the programming languages C and C++.
In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.
The Arduino project started in 2003 as a program for students at the Interaction Design Institute Ivrea in Ivrea, Italy, aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats, and motion detectors.

CHAPTER – 3

HARDWARE COMPONENTS



ARDUINO UNO

Features of the Arduino UNO:

Microcontroller: ATmega328
Operating Voltage: 5V
Input Voltage (recommended): 7-12V
Input Voltage (limits): 6-20V
Digital I/O Pins: 14 (of which 6 provide PWM output)

Analog Input Pins: 6
DC Current per I/O Pin: 40 mA
DC Current for 3.3V Pin: 50 mA
Flash Memory: 32 KB of which 0.5 KB used by bootloader
SRAM: 2 KB (ATmega328)
EEPROM: 1 KB (ATmega328)
Clock Speed: 16 MHz

The name Arduino comes from a bar in Ivrea, Italy, where some of the founders of the project used to meet. The bar was named after Arduin of Ivrea, who was the margrave of the March of Ivrea and King of Italy from 1002 to 1014.

ARDUINO HARDWARE PART:-

Arduino is open-source hardware. The hardware reference designs are distributed under a Creative Commons Attribution Share-Alike 2.5 license and are available on the Arduino website. Layout and production files for some versions of the hardware are also available.

Although the hardware and software designs are freely available under copyleft licenses, the developers have requested the name Arduinoto be exclusive to the official product and not be used for derived works without permission.
The official policy document on use of the Arduino name emphasizes that the project is open to incorporating work by others into the official product. Several Arduinocompatible products commercially released have avoided the project name by using various names ending in -duino.
Most Arduino boards consist of an Atmel 8-bit AVR microcontroller (ATmega8, ATmega168, ATmega328, ATmega1280, ATmega2560) with varying amounts of flash memory, pins, and features.
The 32-bit Arduino Due, based on the Atmel SAM3X8E was introduced in 2012. The boards use single or double-row pins or female headers that facilitate connections for programming and incorporation into other circuits. These may connect with add-on modules termed shields.

Multiple and possibly stacked shields may be individually addressable via an I²C serial bus. Most boards include a 5 V linear regulator and a 16 MHz crystal oscillator or ceramic resonator.
Some designs, such as the LilyPad, run at 8 MHz and dispense with the onboard voltage regulator due to specific form-factor restrictions.
Arduino microcontrollers are pre-programmed with a boot loader that simplifies uploading of programs to the on-chip flash memory. The default bootloader of the Arduino UNO is the optiboot bootloader.
Boards are loaded with program code via a serial connection to another computer. Some serial Arduino boards contain a level shifter circuit to convert between RS232 logic levels and transistor–transistor logic(TTL) level signals.
Current Arduino boards are programmed via Universal Serial Bus (USB), implemented using USB-to-serial adapter chips such as the FTDI FT232. Some boards, such as later-model Uno boards, substitute the FTDI chip with a separate AVR chip containing USB-to-serial firmware, which is reprogrammable via its own ICSP header.

Other variants, such as the Arduino Mini and the unofficial Boarduino, use a detachable USB-to-serial adapter board or cable, Bluetooth or other methods. When used with traditional microcontroller tools, instead of the Arduino IDE, standard AVR in-system programming (ISP) programming is used.

The Arduino board exposes most of the microcontroller's I/O pins for use by other circuits. The Diecimila, Duemilanove, and current Uno provide 14 digital I/O pins, six of which can produce pulse-width modulated signals, and six analog inputs, which can also be used as six digital I/O pins.
These pins are on the top of the board, via female 0.1-inch (2.54 mm) headers. Several plug-in application shields are also commercially available.
The Arduino Nano, and Arduino-compatible Bare Bones Board and Boarduino boards may provide male header pins on the underside of the board that can plug into solderless breadboards.
Many Arduino-compatible and Arduino-derived boards exist. Some are functionally equivalent to an Arduino and can be used interchangeably. Many enhance the basic Arduino by adding output drivers, often for use in school-level education, to simplify making buggies and small robots.

Others are electrically equivalent but change the form factor, sometimes retaining compatibility with shields, sometimes not. Some variants use different processors, of varying compatibility.

Flame sensor :

Flame sensor is the most sensitive to ordinary light that is why its reaction is generally used as flame alarm purposes. This module can detect flame or wavelength in760 nm to 1100 nm range of light source. Small plate output interface can and single-chip can be directly connected to the microcomputer IO port. The sensor and flame should keep a certain distance to avoid high temperature damage to the sensor. The shortest test distance is 80 cm, if the flame is bigger, test it with farther distance.The detection angle is 60 degrees so the flame spectrum is especially sensitive.The detection angle is 60 degrees so the flame spectrum is especially sensitive.

Specifications:

- On-board LM393 voltage comparator chip and infrared sensing probe.
- Support 5V/3.3V voltage input.
- On-board signal output indication, output effective signal is high level, and the same time the indicator light up, output signal can directly connect with microcontroller IO.
- Signal detection sensitivity can be no adjusted.
- Reserved a line voltage compare circuit (P3 is leaded out)
- PCB size: 30(mm) x15(mm).

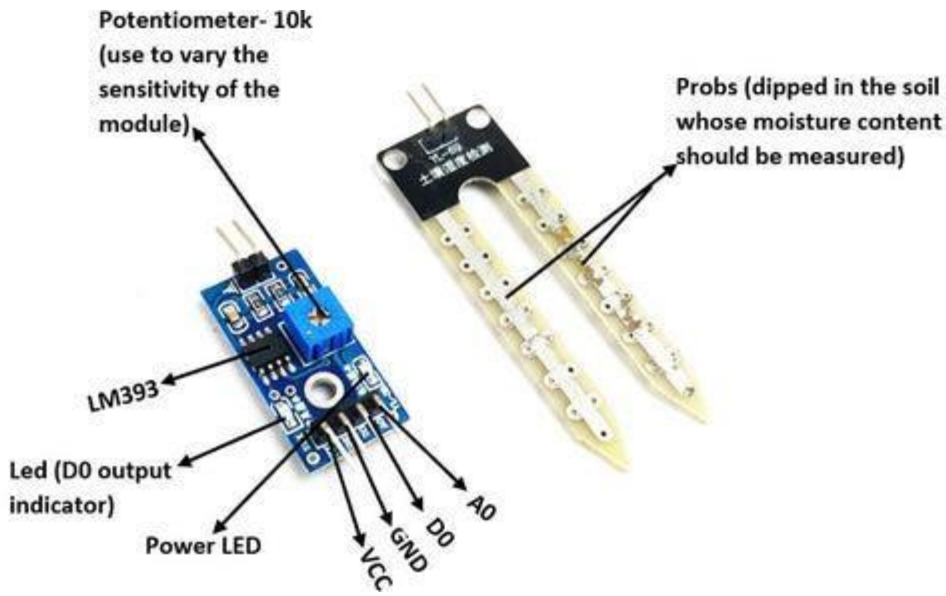
Pin Configuration:

1. VCC
2. Output
3. Ground

Soil moisture sensor :

This soil moisture sensor module is used to detect the moisture of the soil. It measures the volumetric content of water inside the soil and gives us the moisture level as output.

The module has both digital and analog outputs and a potentiometer to adjust the threshold level.



Soil Moisture Sensor Module Pinout Configuration :

Pin Name : Description

VCC : The Vcc pin powers the module, typically with +5V

GND : Power Supply Ground

DO : Digital Out Pin for Digital Output.

AO : Analog Out Pin for Analog Output

Soil Moisture Sensor Module Features & Specifications :

•Operating Voltage: 3.3V to 5V DC
•Operating Current: 15mA
•Output Digital - 0V to 5V, •Adjustable trigger level from preset
•Output Analog - 0V to 5V based on infrared radiation from fire flame falling on the sensor

•LEDs indicating output and power
•PCB Size: 3.2cm x 1.4cm
•LM393 based design

•Easy to use with Microcontrollers or even with normal Digital/Analog IC
•Small, cheap and easily available

Moisture Sensor :

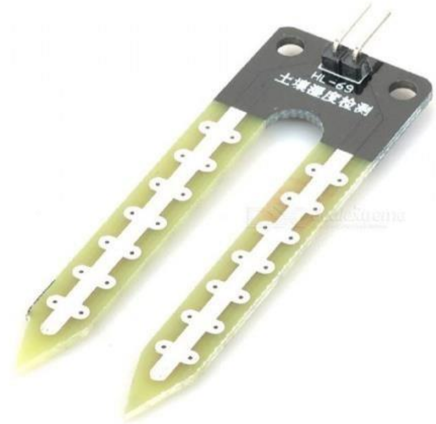
The moisture sensor consists of two probes that are used to detect the moisture of the soil. The moisture sensor probes are coated with immersion gold that protects Nickel from oxidation. These two probes are used to pass the current through the soil and then the sensor reads the resistance to get the moisture values.

LM393 IC :

LM393 Comparator IC is used as a voltage comparator in this Moisture sensor module. Pin 2 of LM393 is connected to Preset (10KΩ Pot) while pin 3 is connected to Moisture sensor pin. The comparator IC will compare the threshold voltage set using the preset (pin2) and the sensor pin (pin3).

Moisture Sensor Probe :

The Moisture Sensor Probe is a fork-like physical package that easily inserts into soil or other soft, non-corrosive media. This sensor uses conductivity to detect the amount of moisture between the probes. The likely use of this sensor is in potted plants, micro-farming research and with some non-soil environments.



Applications of Soil Moisture Sensor :

•Gardening
•Irrigation Systems
•Used in Controlled environments

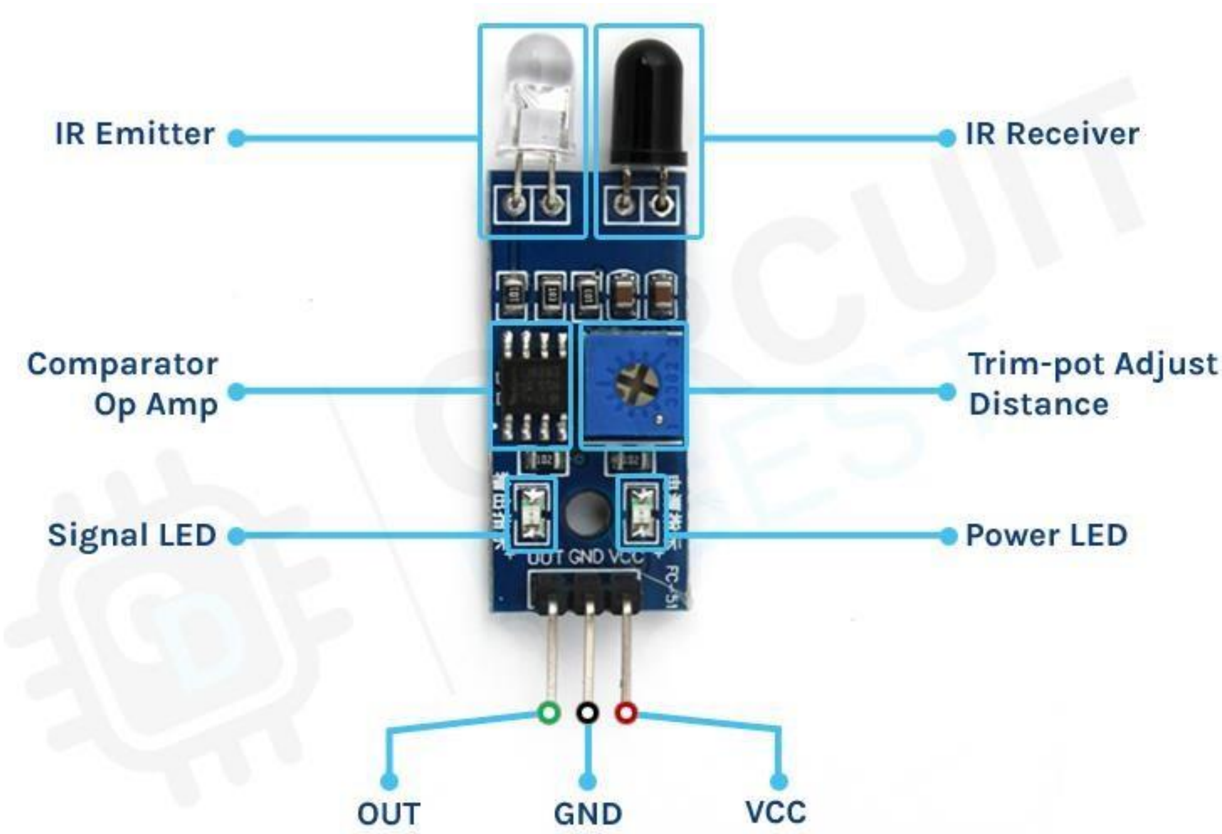
IR sensor :

The IR Sensor Module or infrared (IR) sensor is a basic and most popular sensor in electronics. It is used in wireless technology like remote controlling functions and detection of surrounding objects/ obstacles.
IR sensors mainly consist of an Infrared(IR) LED and a Photodiode, this pair is generally called IR pair. An IR LED is a special purpose LED, it is can emitting infrared rays ranging from 700 nm to 1 mm wavelength.
These types of rays are invisible to our eyes. In contrast, a photodiode or IR Receiver LED detects the infrared rays.

The IR sensor module consists mainly of the

1. IR Transmitter.

2. Photodiode Receiver.
3. LM393 Comparators IC
4. Variable Resistor (Trim pot), 5. Power LED,
6. output LED.



IR sensor module pinout configuration :

1. VCC : +5 v power supply
2. GND : Ground (-) power supply
3. OUT : Digital Output

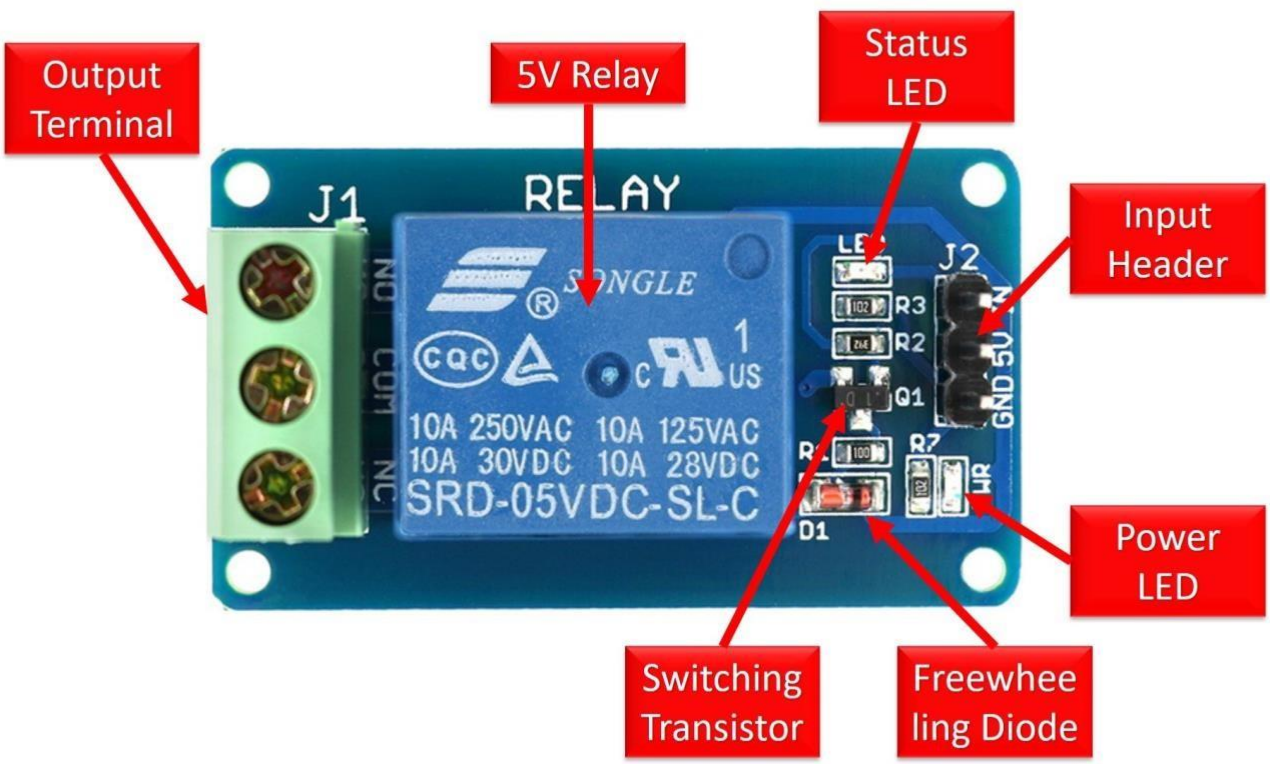
Relay :

A Relay is a simple electromechanical switch. While we use normal switches to close or open a circuit manually, a Relay is also a switch that connects or disconnects two circuits.

But instead of a manual operation, a relay uses an electrical signal to control an electromagnet, which in turn connects or disconnects another circuit.

A relay is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, such as solid-state relays. Relays are used where it is necessary to control a circuit by a separate low-power signal, or where several circuits must be controlled by one signal.

The first relays were used in long distance telegraph circuits as amplifiers: they repeated the signal coming in from one circuit and re-transmitted it on another circuit.



Relays were used extensively in telephone exchanges and early computers to perform logical operations.

A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor. Solid-state relayscontrol power

circuits with no moving parts, instead using a semiconductor device to perform switching.
Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from overload or faults; in modern electric power systems these functions are performed by digital instruments still called "protective relays".
Magnetic latching relays require one pulse of coil power to move their contacts in one direction, and another, redirected pulse to move them back. Repeated pulses from the same input have no effect. Magnetic latching relays are useful in applications where interrupted power should not be able to transition the contacts.
Magnetic latching relays can have either single or dual coils. On a single coil device, the relay will operate in one direction when power is applied with one polarity, and will reset when the polarity is reversed.

On a dual coil device, when polarized voltage is applied to the reset coil the contacts will transition.
AC controlled magnetic latch relays have single coils that employ steering diodes to differentiate between operate and reset commands.

The Arduino Relay module allows a wide range of microcontroller such as Arduino, AVR ,PIC, ARM with digital outputs to control larger loads and devices like AC or DC Motors, electromagnets, solenoids, and incandescent light bulbs.
This module is designed to be integrated with 2 relays that it is capable of control 2 relays.The relay shield use one QIANJI JQC-3F high-quality relay with rated load 7A/240VAC,10A/125VAC,10A/28VDC.
The relay output state is individually indicated by a light-emitting diode.

2 channel relay features:

Number of Relays: 2
Control signal: TTL level
Rated load: 7A/240VAC 10A/125VAC 10A/28VDC
Contact action time: 10ms/5ms

Types of Relays :

Relays can be classified into different types depending on their functionality, structure, application etc. We listed out some of the common types of relays here.

•Electromagnetic
•Latching
•Electronic
•Non-Latching
•Reed
•High-Voltage
•Small Signal
•Time Delay
•Multi-Dimensional

•Thermal
•Differential
•Distance
•Automotive
•Frequency
•Polarized
•Rotary
•Sequence
•Moving Coil
•Buchholz
•Safety
•Supervision
•Ground Fault

Relay Applications :

Relays are used to protect the electrical system and to minimize the damage to the equipment connected in the system due to over currents/voltages. The relay is used for the purpose of protection of the equipment connected with it.

These are used to control the high voltage circuit with low voltage signal in applications audio amplifiers and some types of modems.
These are used to control a high current circuit by a low current signal in the applications like starter solenoid in automobile. These can detect and isolate the faults that occurred in power transmission and distribution system. Typical application areas of the relays include

•Lighting control systems
•Telecommunication
•Industrial process controllers
•Traffic control
•Motor drives control
•Protection systems of electrical power system
•Computer interfaces
•Automotive
•Home appliances

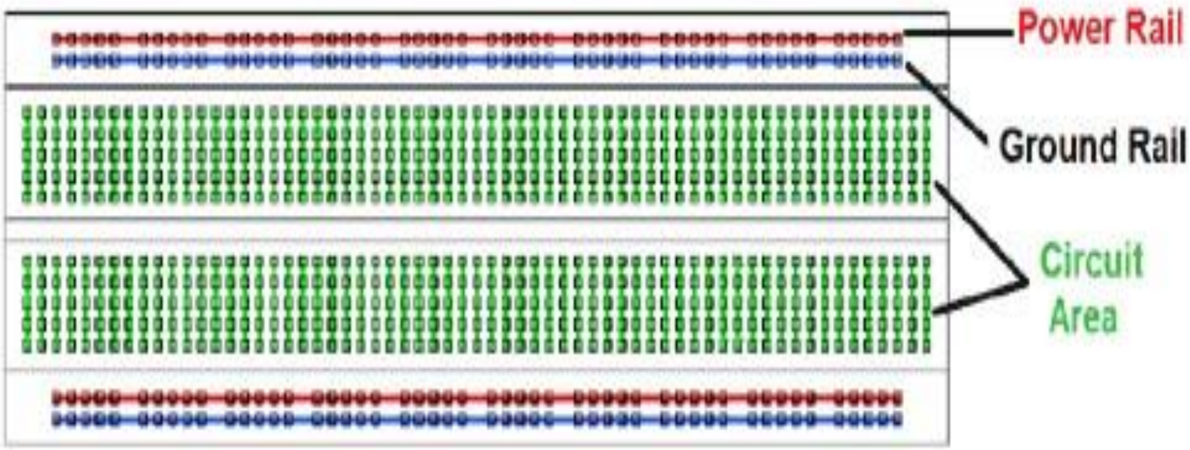
Breadboard :

What is a Breadboard?

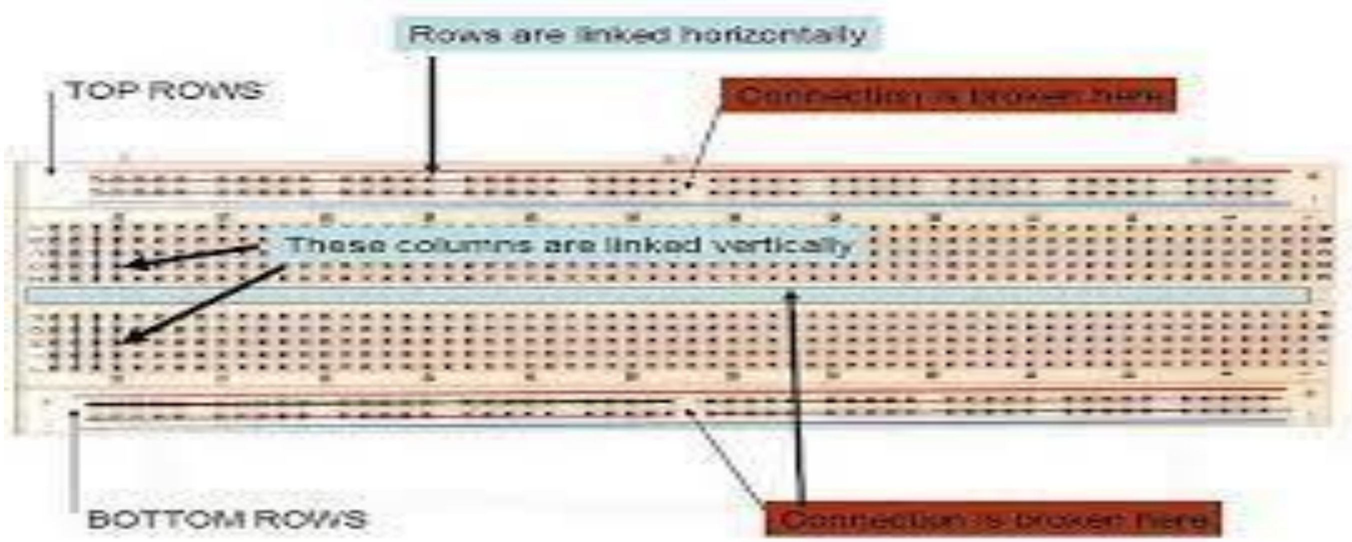
As the name suggests, the term breadboard can be derived from two terms namely bread & board. Initially, this was used to cut the bread into pieces. Further, it was called a breadboard & it was used in electronics projects and electronic devices in the year 1970. A breadboard is also known as a solderless board because the component used on the breadboard does not need any soldering to connect to the board, so it can be reused.

The arrangement of different components on a breadboard can be done by inserting their terminals into the breadboard, so it is frequently known as a plugboard. Breadboard definition is a plastic board in rectangular shape that includes a lot of small holes in it to allow you to place different components to build an electronic circuit is known as a breadboard.

The connection on the breadboard is not permanent but they can be connected without soldering the components If you make any mistake while connecting the components, you can place or remove the components effortlessly. For beginners of electronics, this device is very helpful to make mini-projects. If a designer builds a simple circuit that they desire to analyze, then a breadboard gives a quick solution. The breadboard diagram is shown below.



The material used to make the breadboard is white plastic. At present, most of the breadboards are solderless types, so we can directly plug in the components directly and connected them through the exterior power supply. The different kinds of breadboards are accessible according to the specific point holes. For instance 400 point type, 830 point type, etc.

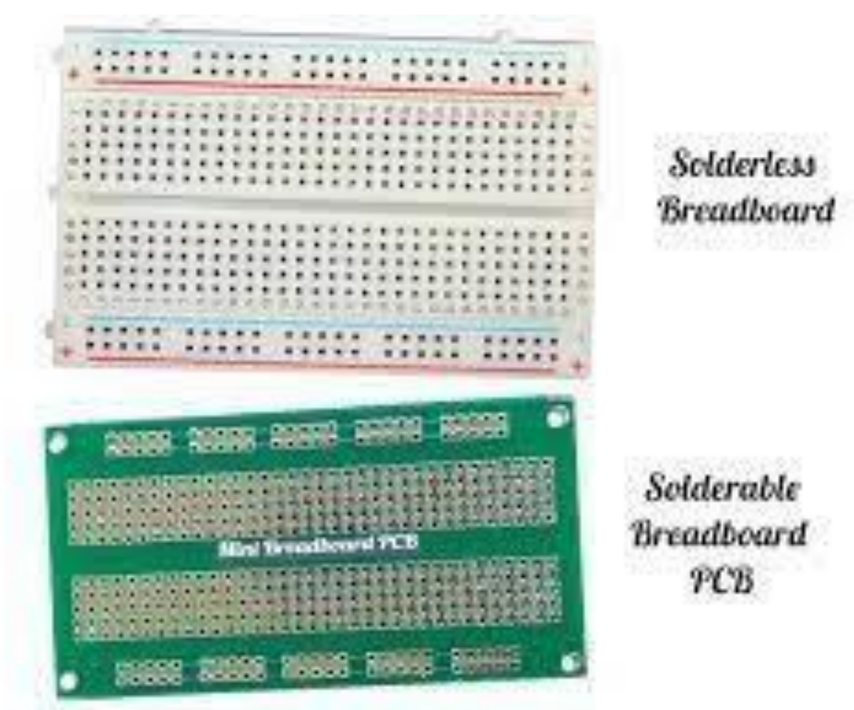


Specifications & Features :

The specifications & features of a breadboard, include the following.

•Distribution Strips are two
•Wire Size is 21 to 26 •AWG wire
•Tie Points are two hundred
•Withstanding in Voltage is 1,000V AC
•Tie points within IC are 630
•Insulation Resistance is DC500V or 500MΩ
•Dimension is 6.5*4.4*0.3 inch
•Rating is 5Amps
•ABS plastic through color legend
•ABS heat Distortion Temperature is 183° F (84° C)Hole or Pitch Style is 2.54mm

Types of Breadboard :



•Solder-less breadboard

•Solder-able breadboard

Solderless Breadboards :

This is the most commonly used breadboard for prototyping as well as testing electronic circuits without soldering the components. These are available in different shapes, sizes as well as ratings.

The circuits on these breadboards are not permanent so we can check & test the functionality of a circuit before confirming its design onto a PCB. These breadboards include rows & columns with holes that allow the leads of components & wire gauges.
If the terminal of the component does not place into the hole of a breadboard, then a connecting wire can be soldered to the lead of a component that will insert in the breadboard hole.
Advantages :

The advantages of solderless breadboards include the following : 1.
It doesn't require soldering to connect the components on board.
2. If the circuit is not working properly then, we can easily check and rectify them by taken out the components & replace them easily.

Disadvantages :

The disadvantages of solderless breadboards include the following :

1. Components that are connected to the breadboard can come loose once the breadboard is pushed or moved.
2. This kind of breadboard is available with high parasitic capacitances because of the capacitances among different components which are being close to each other.
3. These breadboards are restricted to below or 10 MHz frequencies.

Solderable Breadboards :

These types of breadboards offer a permanent setup for your electronic circuits. This kind of breadboard gives a stronger setup. It includes holes for electronic components

including copper tracing. These components can be soldered using soldering iron for soldering the components to the breadboard so that an electrical connection can be formed through the copper tracing.
For designing a circuit, jumper wires are needed for soldering separately in between these components to make a lane to permit the flow of current. These types of breadboards are available in different sizes based on the requirement.

Advantages :

The advantages of solderable breadboards include the following:

1. These breadboards are robust and your circuit will be very secured on this type of breadboard.
2. This kind of breadboard gives your project a more specialized look.
3. Less cost and saves time while designing a circuit.

Disadvantages :

The disadvantages of solderable breadboards include the following :

1. If there is any error occurs in the circuit then de-soldering may cause damage to components
2. This board cannot be reused.

What Are Jumper Wires?

Generally, jumpers are tiny metal connectors used to close or open a circuit part. They have two or more connection points, which regulate an electrical circuit board.

Their function is to configure the settings for computer peripherals, like the motherboard. Suppose your motherboard supported intrusion detection. A jumper can be set to enable or disable it.

Jumper wires are electrical wires with connector pins at each end. They are used to connect two points in a circuit without soldering.

You can use jumper wires to modify a circuit or diagnose problems in a circuit. Further, they are best used to bypass a part of the circuit that does not contain a resistor and is suspected to be bad.

This includes a stretch of wire or a switch. Suppose all the fuses are good and the component is not receiving power; find the circuit switch. Then, bypass the switch with the jumper wire.

How much current (I) and voltage (V) can jumper wires handle?

The I and V rating will depend on the copper or aluminium content present in the wire.

For an Arduino application is no more than 2A and 250V. We also recommend using solid-core wire, ideally 22 American Wire Gauge (AWG).

Jumper Wire Colours :

Although jumper wires come in a variety of colours, they do not actually mean anything. The wire colour is just an aid to help you keep track of what is connected to which.

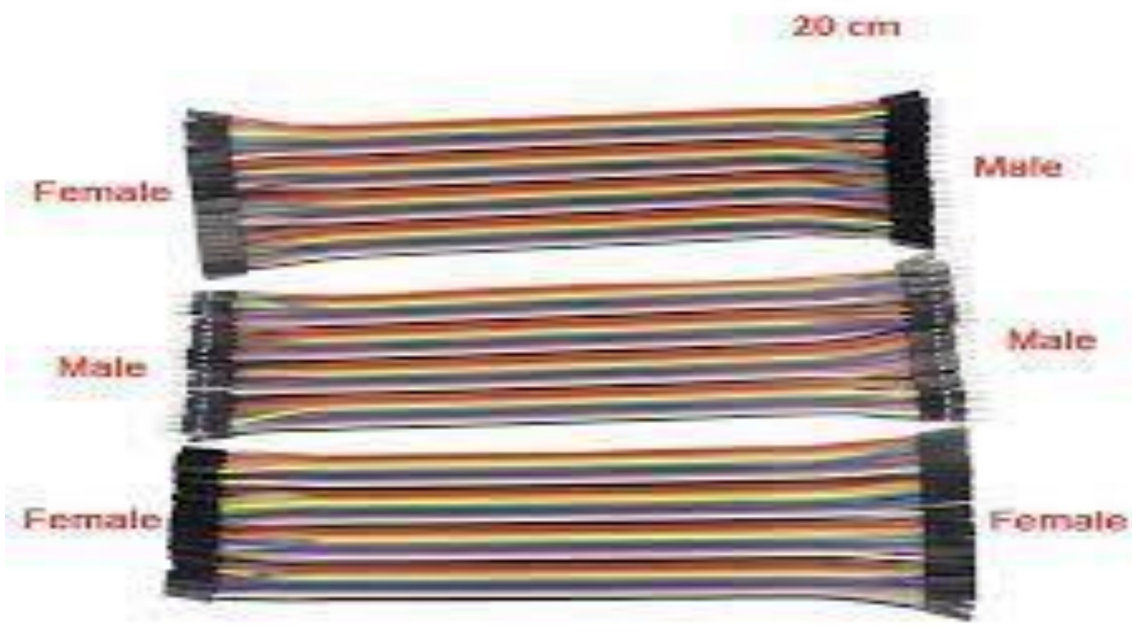
It will not affect the operation of the circuit. This means that a red jumper wire is technically the same as the black one.

Even so, the colours can be used to your advantage to differentiate the types of connections. For instance, red as ground and black as power. Literally, what works for you!

Types of Jumper Wires :

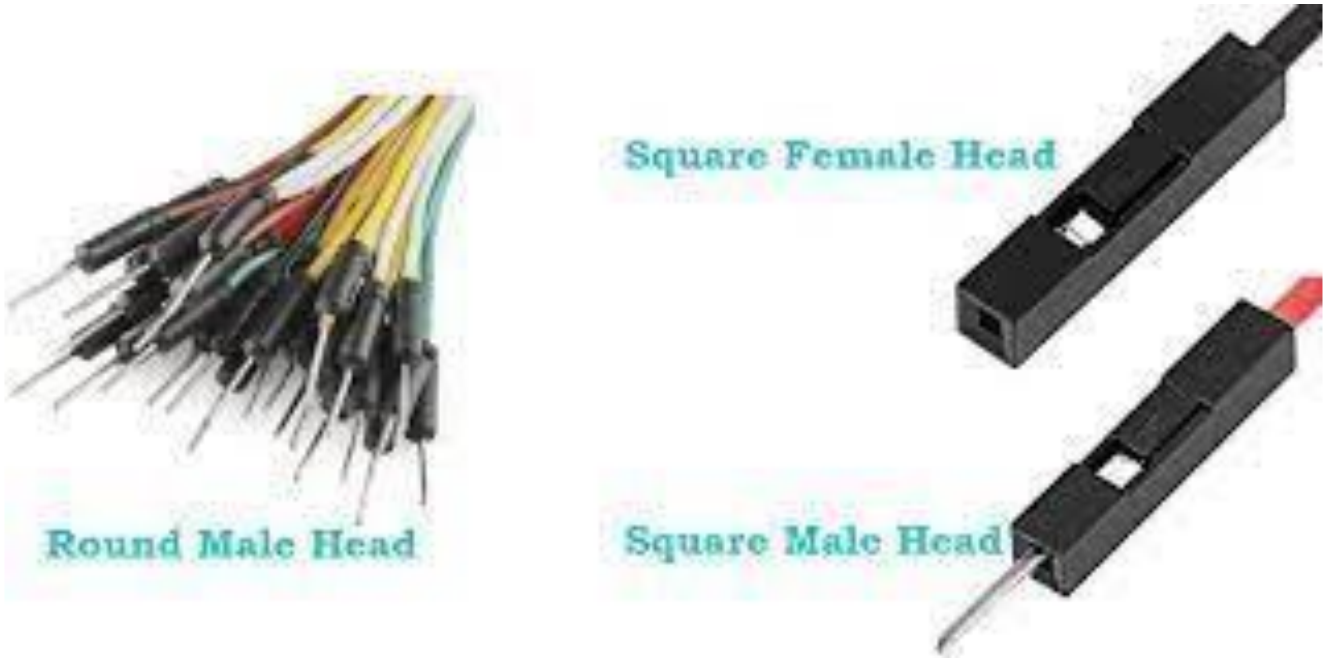
Jumper wires come in three versions:

• Male-to-male jumper
• Male-to-female jumper
• Female-to-female jumper



And two types of head shapes:

1. square head and 2.
round head.



The difference between each is in the endpoint of the wire. Male ends have a pin protruding and can plug into things, while female ends do not but are also used for plugging.

Moreover, a male connector is referred to as a plug and has a solid pin for centre conduction. Meanwhile, a female connector is referred to as a jack and has a centre conductor with a hole in it to accept the male pin.

Male-to-male jumper wires are the most common and what you will likely use most often. For instance, when connecting two ports on a breadboard, a male-to-male wire is what you will need.

How to choose a jumper wire?

This also includes the head shape, though it may depend on the head type you have.

For instance, you are connecting an Arduino Uno pin with a breadboard. A male-tomale jumper is recommended. But if you are connecting it directly to an Ultrasonic Sensor, go for a male-to-female jumper.

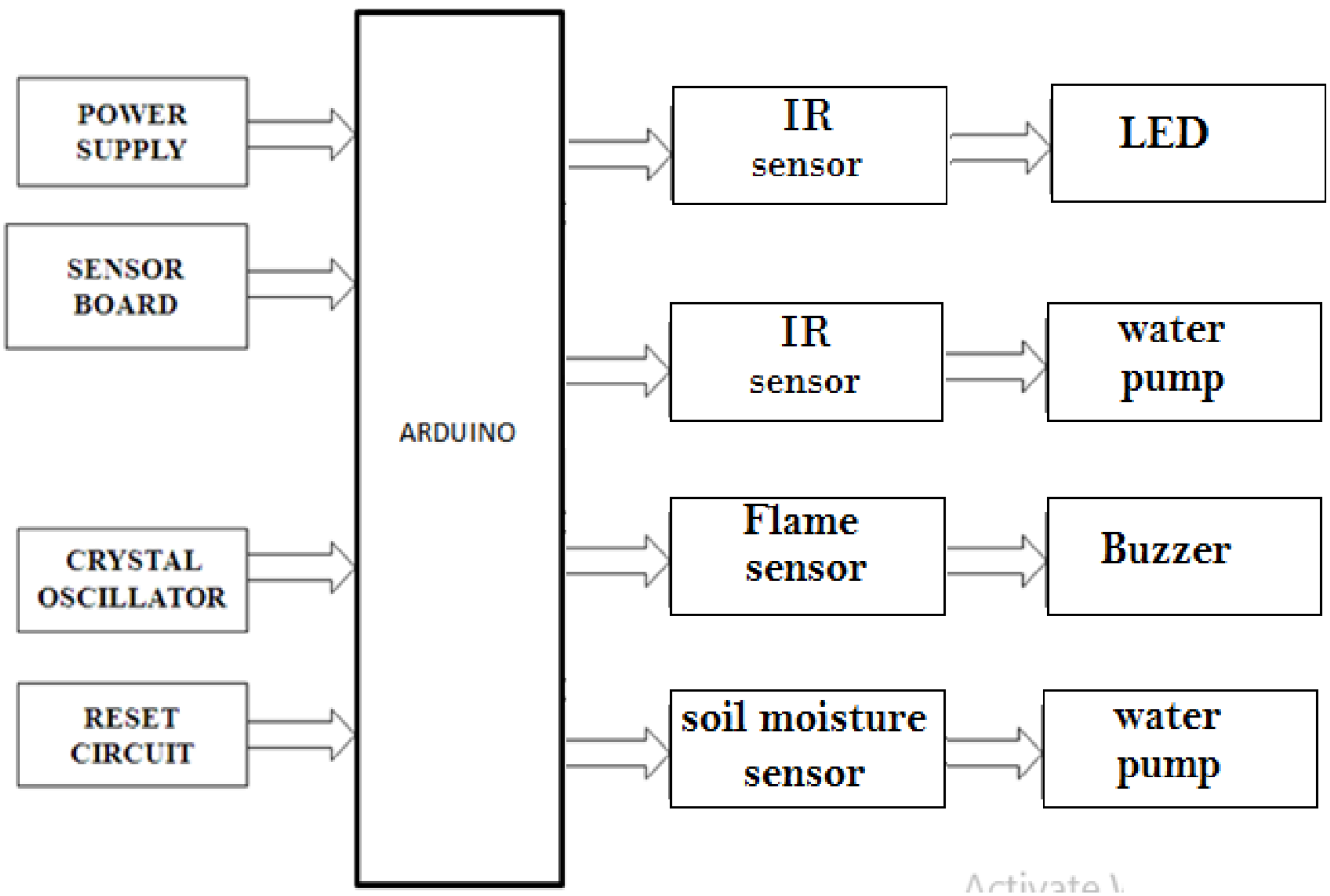
How do you solder a jumper?

First step: Strip the wire, then open the strippers.
Second step: Tin the wire.
Third step: Solder the jumper to the board. Poke your tinned jumper end through your PC board.
Fourth step: Clip the excess. Set your cutters close to the base of the wire. **Fifth step:** De-solder the joint.

CHAPTER – 4

BLOCK DIAGRAM

BLOCK DIAGRAM



ARDUINO SOFTWARE PART:-

IDE

The Arduino integrated development environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in the programming language Java. It originated from the IDE for the languages Processing and Wiring. It includes a code editor with features such as text cutting and pasting, searching and replacing text, automatic indenting, brace matching, and syntax highlighting, and provides simple one-click mechanisms to compile and upload programs to an Arduino board. It also contains a message area, a text console, a toolbar with buttons for common functions and a hierarchy of operation menus. The source code for the IDE is released under the GNU General Public License, version 2.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that

are compiled and linked with a program stub main() into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.

CHAPTER – 5

CODE

CODE

```
int IRSensor = 5; // connect ir sensor to arduino pin 5 int
soilSensor = 8; //sensor digital pin vonnected to pin 8 int
flameSensor = 4; // connect flame sensor to arduino pin 4
int val; //This variable stores the value received from Soil moisture sensor
int WATERPUMP1 = A1; //motor pump connected to pin A1
int WATERPUMP2 = A2; //motor pump connected to pin
A2 int waterpump3 =A3;//motot pump connected to pin A3
int buzzer=11;// ALARAM
int led =9;// the pin that the LED is connected 9 int
IRSensor1 = 7;// the pin that the sensor is connected 7
int state = LOW;// by default, no motion detected
int val1 = 0;// variable to store the sensor status (value)int buzzer = 10;
int smokeA0 = 3; // Your threshold value
//int sensorThres = 450;


void setup() {
pinMode(A1,OUTPUT); //Set pin A1 as OUTPUT pin pinMode(6,INPUT);
//Set pin 6 as input pin, to receive data from Soil moisture sensor.
pinMode (IRSensor, INPUT); // sensor pin INPUT
pinMode (A2, OUTPUT); // WATERPUMP2 OUTPUT
pinMode (flameSensor, INPUT); // sensor pin INPUT pinMode
(11, OUTPUT); // ALARAM OUTPUT
pinMode(10,INPUT);   pinMode   (A3,   OUTPUT);   //
WATERPUMP3 OUTPUT
pinMode(led, OUTPUT);      // initalize LED as an output pinMode(IRSensor1,
INPUT);    // initialize sensor as an input
pinMode(smokeA0, INPUT);
Serial.begin(9600);
}          void
loop(){
  smoke();
light();   plants();
fire();
  toilet();
```

```
}

void light(){
  int statusSensor2 = digitalRead (IRSensor1);

  if (statusSensor2 == 1)
  {
    digitalWrite(9, LOW); // LED LOW
  }
else
{
    digitalWrite(9, HIGH); // LED High
    //delay(A3000);
  }

}
void plants(){
   val = digitalRead(8);  //Read data from soil moisture sensor     if(val
== HIGH)
  {
  digitalWrite(A1,HIGH); //if soil moisture sensor provides LOW value send LOW
value to motor pump and motor pump goes off
  }
else
  {
  digitalWrite(A1,LOW); //if soil moisture sensor provides HIGH value send HIGH
value to motor pump and motor pump get on
  }
 // delay(400); //Wait for few second and then continue the loop.
}        void
fire(){
   int statusSensor1 = digitalRead (flameSensor);

  if (statusSensor1== 1)
  {
    digitalWrite(11, LOW); // LED LOW
```

```
  }
else
  {
    digitalWrite(11, HIGH); // LED High
    delay(3000);
 } } void
toilet(){

  int statusSensor = digitalRead (IRSensor);

  if (statusSensor == 0)
  {
    digitalWrite(A2, HIGH); // LED LOW
    //delay(2000);
  }
    if(statusSensor ==1)
    {
    digitalWrite(A2,LOW); // LED High
    }
  }
  void smoke() {
      int  statusSensor3  =  digitalRead  (smokeA0);
if (statusSensor3 ==LOW)
      {
        digitalWrite(11,HIGH);
      }
else
      {
        digitalWrite(11,LOW);
      }
  }
```

CHAPTER – 6

CONCLUSION

CONCLUSION

Hence we conclude that the required goals and objectives of smart college using arduino has been achieved .

Lights can be controlled to ON and OFF automatically with out any human interference in operating the appliances , by using automatic toilet flush toilets can be flushed in a proper way, so that we can over come the problem of getting bad smell from toilets.

when any fire accidents take place fire alarm detects the fire and gives us a buzzer sound , so that all the students and teachers can get an alert signal which helps them to take certain safety measures.

When we come to the watering of plants it seem difficult to water those plants ,so here by using soil moisture sensor plants can be watered automatically whenever the soil needs watering.

In this way we can reduce the efforts of humans and can make the works easy and simple to do.

Finally this proposed system is better from the scalability and flexibility point of view.